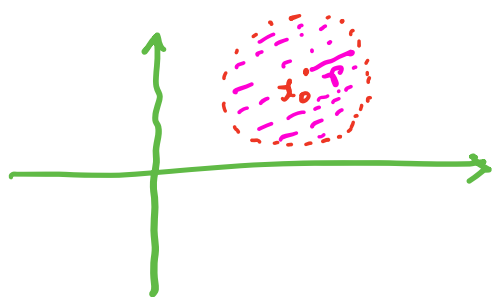


• Elements of Point Set Topology on $\mathbb{R}^n$:

• Open disk in $\mathbb{R}^2$:

Let $x_0 \in \mathbb{R}^2$, $r > 0$

The open disk of radius r
 centered at $x_0 = B_r(x_0) = \{x \in \mathbb{R}^2 : \|x - x_0\| < r\}$
 $\subseteq \mathbb{R}^2$



• In $\mathbb{R}$, $B_r(x_0) = \text{open interval } (x_0 - r, x_0 + r)$
 $= \{x \in \mathbb{R} : |x - x_0| < r\}$

So, $B_2(5) = \{x \in \mathbb{R} : |x - 5| < 2\} = (3, 7)$

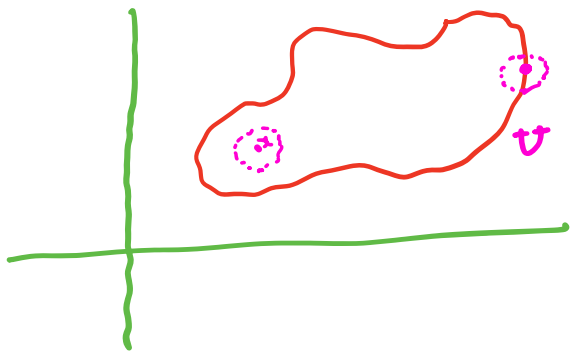


In $\mathbb{R}^n$:

The open ball of radius r centered at x_0 :

$$B_r(x_0) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\}$$

Interior point of a Set



Let $U \subseteq \mathbb{R}^n$

A point $x \in U$ is called an interior point of U

if there is some open ball with center x lying entirely in U .

A point $x \in \mathbb{R}^n$ is called a boundary point of U if every open ball with center x contains points of U as well as points outside of U .

(ie. for all $r > 0$, $B_r(x) \cap U \neq \emptyset$
and $B_r(x) \cap U^c \neq \emptyset$)

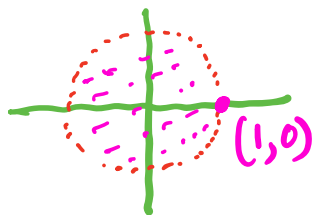
Let $U \subseteq \mathbb{R}^n$:

- Interior of a set U (denoted by $\text{int}(U)$ or U°) is the set of all interior points of U .
- Boundary of a set U (denoted by ∂U) is the set of all boundary points of U .

Now $\text{int}(U) \subseteq U$

Question: What about ∂U ?

Ex: Let $A = B_1((0,0)) \cup \{(1,0)\}$



$$\text{Then } \text{int}(A) = B_1((0,0))$$

$$\text{and } \partial A = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

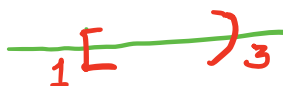
Open set:

A set $U \subseteq \mathbb{R}^n$ is called open if $\boxed{U = \text{int}(U)}$ (i.e. it consists of entirely its interior points)

closed set:

A set $U \subseteq \mathbb{R}^n$ is called closed if $\boxed{\partial U \subseteq U}$ (i.e. it contains all its boundary points)

Ex: Let $A = [1, 3)$



$$\text{Then } \partial A = \{1, 3\}$$

$$\text{int } A = (1, 3)$$

A is not closed since $3 \notin A$

A is not open since 1 is not an interior point.

So, $\boxed{[1, 3)}$ is neither open nor closed

But $A = (1, 3)$ is an open set in $\mathbb{R}$. ($\partial A = \{1, 3\}$)

and $A = [1, 3]$ is a closed set in $\mathbb{R}$ ($\partial A = \{1, 3\}$)

and $\boxed{A = (1, 3]}$ is neither open, nor closed ($\partial A = \{1, 3\}$)

Ex: Let $A = (1, 3)$

$$\text{Then } \text{Int}(A) = (1, 3) \quad \left(\text{So, } A \text{ is open} \right)$$

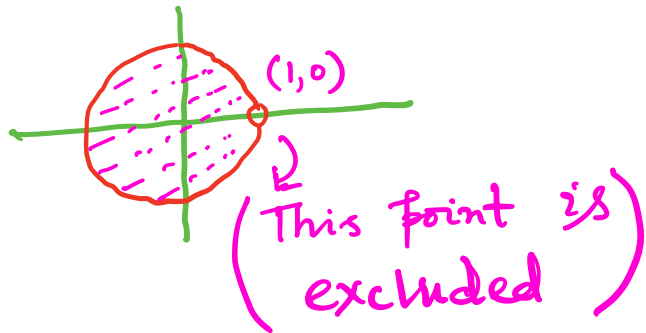
$$\partial A = \{1, 3\}$$

Ex: Let $A = [1, 3]$ Then $\text{Int}(A) = (1, 3)$
 $\partial A = \{1, 3\}$

Ex: Let $A = (1, 2) \cup (2, 3)$ 
 Then $\text{Int} A = (1, 2) \cup (2, 3) = (1, 3) \setminus \{2\}$
 $\partial A = \{1, 2, 3\}$
 A is open

Ex: Let $A = B_1((0,0)) \cup \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} \setminus \{(1,0)\}$

Then $\text{Int}(A) \equiv B_1((0,0))$

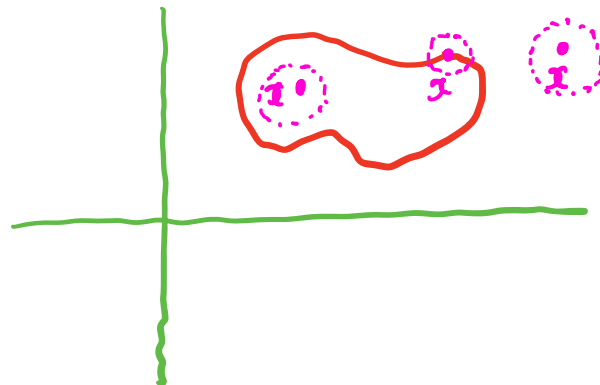


$\partial A = \{(x,y) : x^2 + y^2 = 1\}$

So, A is neither open nor closed.

Let $A \subseteq \mathbb{R}^n$

Now given $x \in \mathbb{R}^n$, one and only one possibility must hold.



• $\exists r > 0$ such that $B_r(x) \subseteq A$
 ($\exists \equiv$ There exists)

→ In this case, x is an interior point of A .

OR
 • $\exists r > 0$ such that $B_r(x) \subseteq A^c$

→ In this case, x is an exterior point of A .

OR
 $\forall r > 0$, we have $B_r(x) \cap A \neq \emptyset$
 and $B_r(x) \cap A^c \neq \emptyset$

→ In this case, x is a boundary point of A

Ex: Find the exterior, interior and boundary of the following sets:

(1) $\{x \in \mathbb{R}^n : \|x\| \leq 1\}$ (2) $\{x \in \mathbb{R}^n : \|x\| = 1\}$

(3) $\{x \in \mathbb{R}^n : \|x\| < 1\}$

(4) $\{x \in \mathbb{R}^n : \text{Each } x_i \text{ is rational}\}$

Ex: Let $A \subseteq \mathbb{R}^n$

When can we say that $A = \text{int}(A) \cup \partial A$?

Ex: Let $U = (3, \infty) \subseteq \mathbb{R}$

What is ∂U ?

Ex: Is $\mathbb{R}^n$, an open subset of $\mathbb{R}^n$?

Is $\mathbb{R}^n$, a closed subset of $\mathbb{R}^n$?

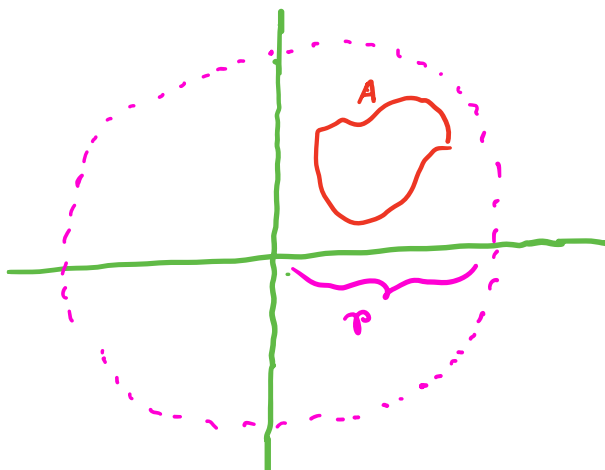
Give examples of proper open and proper closed subsets of $\mathbb{R}^n$.

Bounded set:

Let $U \subseteq \mathbb{R}^n$.

U is called a Bounded Subset of $\mathbb{R}^n$ if there exists some $r > 0$ such that

$$U \subseteq B_r(\mathbf{0}) \text{ where } \mathbf{0} = (0, 0, \dots, 0)$$



Unbounded set:

A set $U \subseteq \mathbb{R}^n$ is called unbounded if it is not bounded.

Ex: Let $f(x, y) = \sqrt{y - x^2}$

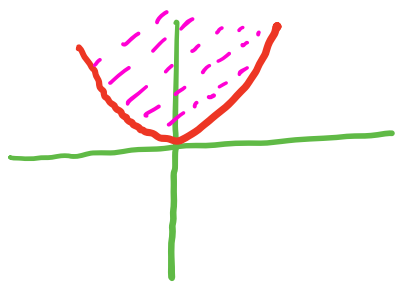
Find the domain of f and range of f .

Determine if the domain is open, closed or neither.

Determine if the domain is bounded or unbounded.

$$\text{Domain} = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$$

$$\text{Range} = [0, \infty)$$

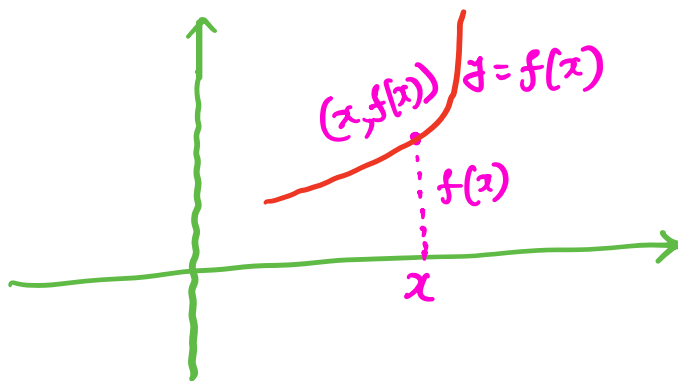


The Domain is closed and unbounded.

$$\text{Boundary of the domain} = \{(x, y) : y = x^2\}$$

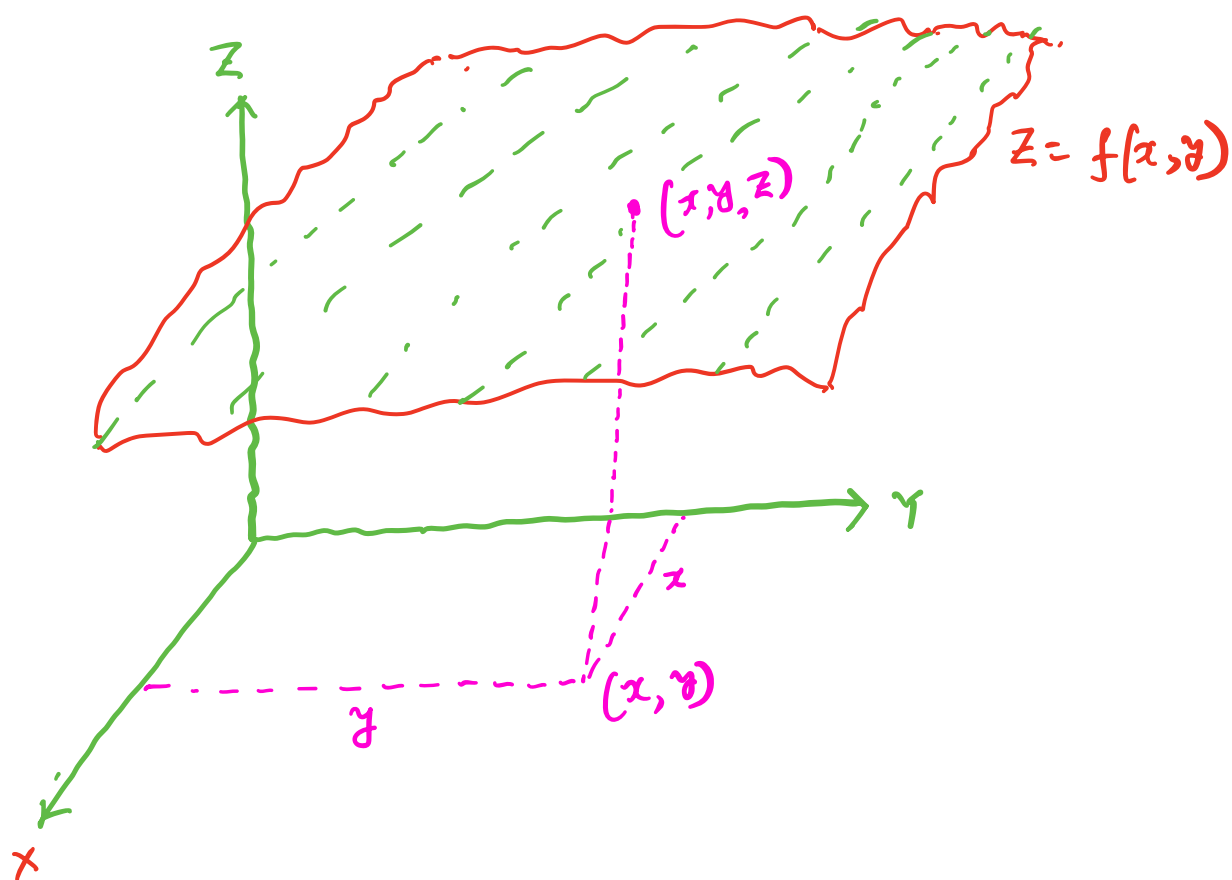
Graph of a function in two variables

First Recall that graph of a function of one variable $y = f(x)$ is a curve in plane.



Now let $z = f(x, y)$ be a function of two variables.

Then the graph of $z = f(x, y)$ represents a surface in space.



Now for n variables

Let $D \subseteq \mathbb{R}^n$ and $f: D \rightarrow \mathbb{R}$ be a function of n variables

The Graph of the function $G(f)$ is defined by

$$G(f) = \{ (x, f(x)) : x \in D \} \subseteq \mathbb{R}^{n+1}$$

$\downarrow$ $\downarrow$
 n tuple Real number

$(n+1)$ tuple with $(n+1)$ coordinates

Level Set:

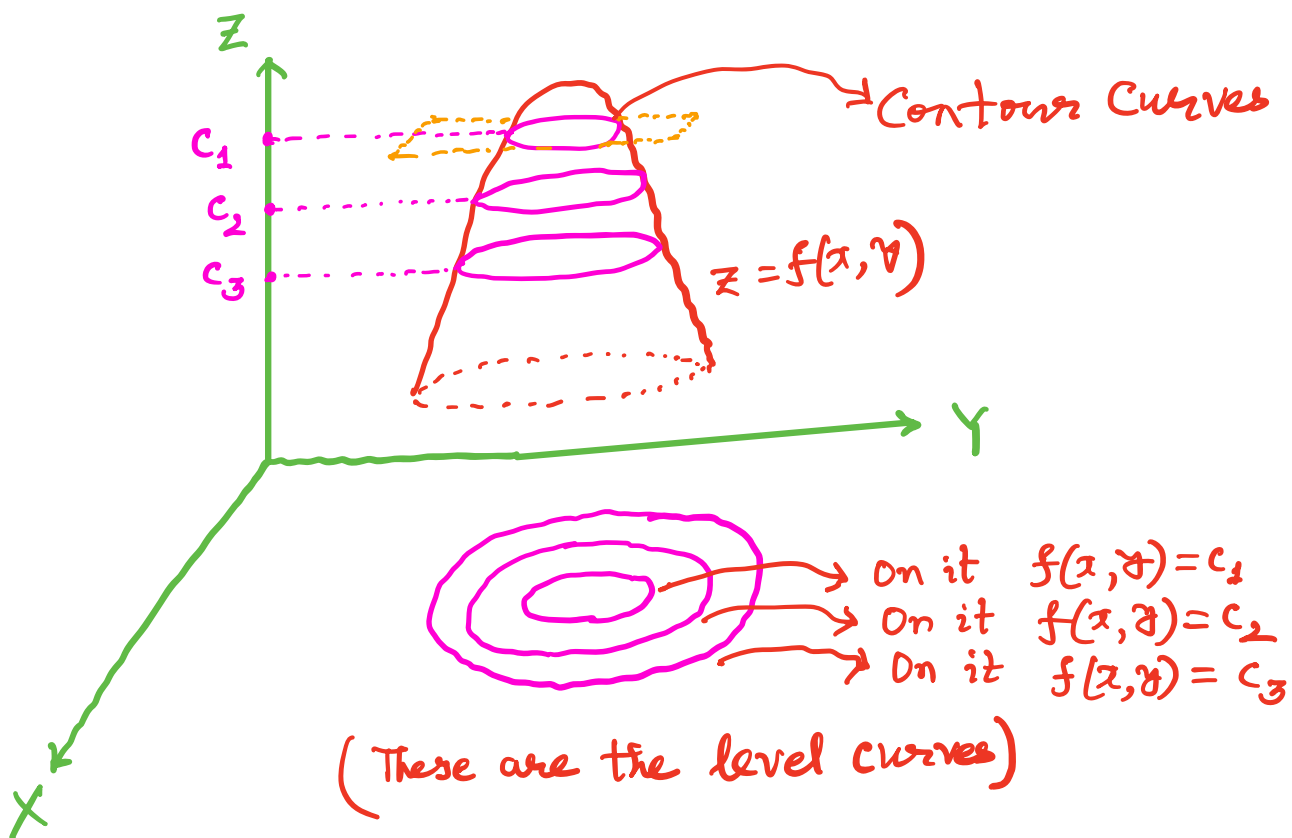
Let n be a positive integer and $D \subseteq \mathbb{R}^n$

Let $f: D \rightarrow \mathbb{R}$ be a function

Given a constant c , the level set of f at c
is defined as $\{x \in D : f(x) = c\} (\subseteq \mathbb{R}^n)$

Note: Every level set lies in the domain of the function. On each level set, the value of the function is a constant.

- When $n=2$, we call it a Level Curve.
- When $n=3$, we call it a Level surface.

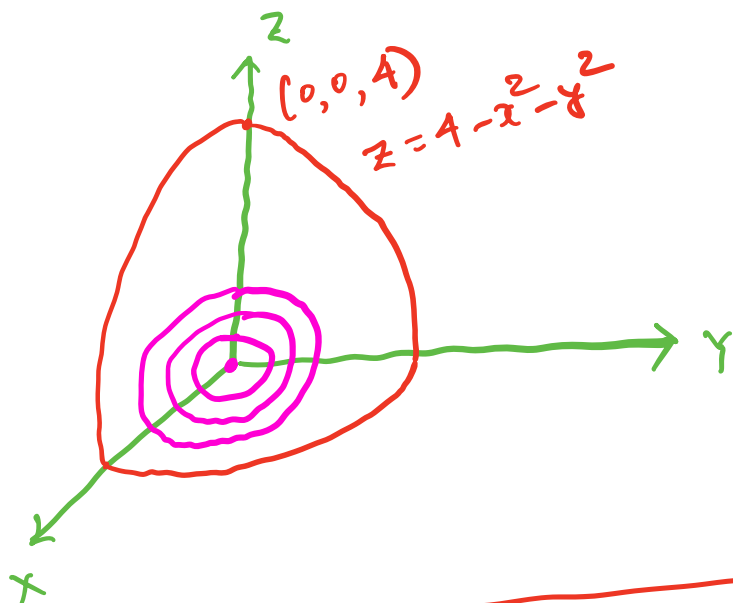


Ex: Let $f(x, y) = 4 - x^2 - y^2$
 Find the level curves.

Let $4 - x^2 - y^2 = c$
 where c is a
 Constant

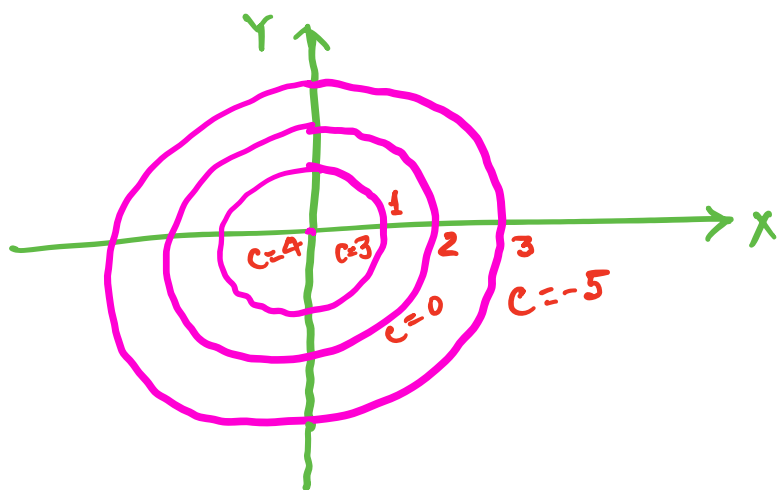
$$\Rightarrow x^2 + y^2 = 4 - c$$

So, it is necessary
 to have $c \leq 4$



The level curves are concentric circles of
 different radius with center at $(0, 0)$

- $c = 4$ gives $x^2 + y^2 = 0$ (circle of radius 0 (just the origin))
 - $c = 3$ gives $x^2 + y^2 = 1$ (circle of radius 1)
 - $c = 0$ gives $x^2 + y^2 = 4$ (circle of radius 2)
- For $c < 0$, we get circles of radius > 2



Ex: Let $f(x, y) = \frac{1}{\sqrt{16 - x^2 - y^2}}$

Find the level curves.

Ex: Find the level surface of
 $f(x, y, z) = x^2 + y^2 + z^2$